shallow interest in literature thanks to the effect of Graham Greene's account of his flirt with such a game; it bore a stronger effect on me than the actual events I had recently witnessed. Greene claimed that he once tried to soothe the dullness of his childhood by pulling the trigger on a revolver – making me shiver at the thought that I had at least a one in six probability of having been without his novels.

The reader can see my unusual notion of alternative accounting: $10 million earned through Russian roulette does not have the same value as $10 million earned through the diligent and artful practice of dentistry. They are the same, can buy the same goods, except that one's dependence on randomness is greater than the other. To an accountant, though, they would be identical. To your next-door neighbor too. Yet, deep down, I cannot help but consider them as qualitatively different. The notion of such alternative accounting has interesting intellectual extensions and lends itself to mathematical formulation, as we will see in the next chapter with our introduction of the Monte Carlo engine. Note that such use of mathematics is only illustrative, aiming at getting the intuition of the point, and should not be interpreted as an engineering issue. In other words, one need not actually compute the alternative histories so much as assess their attributes. Mathematics is not just a "numbers game", it is a way of thinking. We will see that probability is a qualitative subject.

### AN EVEN MORE VICIOUS ROULETTE

Reality is far more vicious than Russian roulette. First, it delivers the fatal bullet rather infrequently, like a revolver that would have hundreds, even thousands of chambers instead of six. After a few dozen tries, one forgets about the existence of a bullet, under a numbing false sense of security. The point is dubbed in this book the *black swan problem*, which we cover in Chapter 7, as it is linked to the problem of induction, a problem that has kept a few philosophers of science awake at night. It is also related to a problem called *denigration of history* as gamblers, investors, and decision makers feel that the sort of things that happen to others would not necessarily happen to them.

Second, unlike a well-defined precise game like Russian roulette, where the risks are visible to anyone capable of multiplying and dividing by six, one does not observe the barrel of reality. Very rarely is the generator visible to the naked eye. One is thus capable of unwittingly playing Russian roulette – and calling it by some alternative "low risk" name. We see the wealth being generated, never the processor, a matter that makes people lose sight of their risks, and never the losers. The game seems terribly easy and we play along blithely.



## *Smooth Peer Relations*

*The degree of resistance to randomness in one's life is an abstract idea,* part of its logic counterintuitive, and, to confuse matters, its realizations non-observable. But I have been increasingly devoted to it – for a collection of personal reasons I will leave for later. Clearly my way of judging matters is probabilistic in nature; it relies on the notion of what could have *probably* happened, and requires a certain mental attitude with respect to one's observations. I do not recommend engaging an accountant in a discussion about such probabilistic considerations. For an accountant a number is a number. If he were interested in probability he would have gotten involved in more introspective professions – and would be inclined to make a costly mistake on your tax return.

While we do not see the roulette barrel of reality, some people give it a try; it takes a special mindset to do so. Having seen hundreds of people enter and exit my profession (characterized by extreme dependence on randomness), I have to say that those who have had a modicum of scientific training tend to go the extra mile. For many, such thinking is second nature. This might not necessarily come from their scientific training *per se* (beware of causality), but possibly from the fact that people who have decided at some point in their lives to devote themselves to scientific research tend to have an ingrained intellectual curiosity and a natural tendency for such introspection. Particularly thoughtful are those who had to abandon scientific studies because of their inability to keep focused on a narrowly defined problem. Without excessive intellectual curiosity it is almost impossible to complete a Ph.D. thesis these days, but without a desire to narrowly specialize it is